

Simple Linear Regression Manual Calculation

Part 1

Simple 2-Point Manual Calculations

Before writing code or utilizing libraries like scikit-learn, it is crucial to understand the underlying data flow. By manually calculating the slope and intercept using just two rows of data, you map the exact logic the algorithm executes internally.

The Dataset (Engine Size vs. CO2)

Observation	X (Engine Size)	Y (CO2 Emissions)
Row 1	2	200
Row 2	4	300

Step-by-Step Execution (Pen & Calculator Workspace)

Step 1: Calculate the Mean of X ($\bar{x}$) and Mean of Y ($\bar{y}$)

Action: Add the X values and divide by the total count. Do the same for Y.

[Workspace for calculation]

Step 2: Find the Deviations: (X - $\bar{x}$) and (Y - $\bar{y}$)

Action: For each row, subtract the mean from the actual value.

[Workspace for calculation]

Step 3: Calculate the Variance of X: $\Sigma(X - \bar{x})^2$

Action: Square each X deviation and sum them together.

[Workspace for calculation]

Step 4: Calculate the Covariance of X and Y: $\Sigma[(X - \bar{x})(Y - \bar{y})]$

Action: Multiply the X deviation by the Y deviation for each row, then sum them together.

[Workspace for calculation]

Step 5: Calculate the Slope (m)

Action: The architecture here is Covariance divided by Variance.

[Workspace for calculation]

Step 6: Calculate the Intercept (c)

Action: Anchor the line using the formula: $c = \bar{y} - (m * \bar{x})$

[Workspace for calculation]

Step 7: The Final Equation

Action: Write your model in the form $y = mx + c$.

[Workspace for calculation]

Answer Key & Verification

Step 1 (Means):

$$\bar{x} = (2 + 4)/2 = 3$$

$$\bar{y} = (200 + 300)/2 = 250$$

Step 2 (Deviations):

$$\text{Row 1: } X_{\text{dev}} = (2 - 3) = -1 \mid Y_{\text{dev}} = (200 - 250) = -50$$

$$\text{Row 2: } X_{\text{dev}} = (4 - 3) = 1 \mid Y_{\text{dev}} = (300 - 250) = 50$$

Step 3 (Variance of X):

$$(-1)^2 + (1)^2 = 1 + 1 = 2$$

Step 4 (Covariance of XY):

$$(-1 * -50) + (1 * 50) = 50 + 50 = 100$$

Step 5 (Slope m):

$$100 / 2 = 50$$

Step 6 (Intercept c):

$$250 - (50 * 3) = 250 - 150 = 100$$

Step 7 (Final Model):

$$y = 50x + 100$$

Architectural Check:

If you plug $X=2$ into your final equation ($y = 50*2 + 100$), you get exactly 200. If you plug $X=4$ in, you get exactly 300. Because we only used two points, our line fits the data perfectly ($R^2 = 1.0$, $MSE = 0$). When you add a third point that doesn't fall exactly on this line, the MSE will rise above zero, and the model will attempt to find the optimal middle ground.